

Approach Note

I used a Siamese ResNet18 model to learn saree design similarity while reducing dependence on colour. Images are resized to 224×224, converted to tensors, and normalized. Training uses paired images with contrastive loss, random horizontal flips, color jitter, and random grayscale augmentation. The network generates 128-D embeddings. During inference, query and gallery embeddings are compared using cosine similarity, followed by ranking for identification or threshold-based verification.

Evaluation

I evaluated the system using two tasks: identification and verification. For identification, I used 115 images as the gallery and 25 images as queries. Each query was converted into a 128-dimensional embedding and compared with all gallery embeddings using cosine similarity. The system achieved 100% Top-1 accuracy and 100% Top-5 accuracy, with all 25 queries retrieving the correct design at rank 1.

For verification, I created 25 positive pairs representing the same/similar design under colour changes and 25 negative pairs containing different images. Cosine similarity was used as the verification score. I selected the decision threshold using the ROC curve and Youden's J statistic, resulting in a threshold of 0.9437. The system achieved 100% accuracy, 100% precision, 100% recall, 100% F1-score, and ROC-AUC of 1.00.

I also evaluated colour robustness separately. The average similarity for the same design under colour transformation was 98.47%, compared with 78.15% for different images. This evaluation was chosen to check both the retrieval ability of the model and its ability to remain consistent when the same design appears with different colours.

Efficiency Report

I used ResNet18 as the backbone because it provides a good balance between feature quality and computational cost compared with larger CNN architectures. The model generates compact 128-dimensional embeddings, which makes gallery storage and cosine-similarity search lightweight. The measured average embedding inference latency was approximately 11.15 ms, with a minimum of 10.73 ms and maximum of 11.77 ms. The gallery contains 115 precomputed embeddings, so inference only requires one query embedding followed by similarity comparison rather than re-processing every gallery image.